Aakansha Kansal

Frontend Developer | React.js & Three.js Engineer

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PROFILE

Frontend Developer with 3+ years building performant, scalable React.js applications, including WebGL-based 3D visualization using Three.js and React Three Fiber. Experienced across the full frontend stack — component architecture, state management, REST/WebSocket integration, and rendering performance — with a specialization in real-time 3D interfaces for enterprise infrastructure platforms. Comfortable owning both standard product UI work and advanced 3D/visualization features within the same codebase.

EDUCATION

B.Tech - Computer Science & Engineering 07/2019 – 07/2023
AKTU, Ghaziabad
CGPA: 8.9

EXPERIENCE

Webforest Agency 09/2026 – Present | Remote (Surat-based team)
Frontend & Three.js Developer

- Building **Badvisor**, a browser-based 3D editor and viewer, in React 19, Three.js and React Three Fiber.
- Implemented **WebGPU rendering with automatic WebGL fallback** and a post-processing effects pipeline.
- Built the asset pipeline for GLB/glTF, OBJ+MTL and VRM models with **Draco and KTX2/Basis compression**, plus Gaussian splat support.
- Designed a camera system with orbit and first-person modes: inertia, auto-rotate, zoom/pan limits, per-device focal lengths and click-to-walk navigation with collision detection.
- Shipped core editor tools: selection, move/rotate/scale gizmos, keyboard shortcuts, drag-and-drop placement and a scene outliner.
- Built a 3D text editor and a material system supporting image, video, HDR and cube-map textures.
- Designed the editor UI in Tailwind CSS: floating property panels, drag-to-scrub number inputs and live-updating fields.

SYSTALDYN CONSULTANCY PVT. LTD. 04/2023 – 09/2026 | Noida
Frontend Developer | React.js & Three.js Engineer

Led the development of enterprise-grade frontend applications and interactive 3D visualization platforms for data center infrastructure monitoring, enabling real-time visualization of power, temperature, alarms, and operational metrics.

Key Achievements

- Architected **DCOS**, a 3D data center digital twin in React and Three.js scaling to **500+ server racks**, with drag-and-drop rack placement and grid-based layout editing.
- Built **multi-user real-time collaboration** using Croquet's publish-subscribe model, keeping shared 3D scene state in sync across users.
- Integrated REST APIs and WebSockets to stream **live power, temperature and alarm data** into 3D scenes and Highcharts dashboards.
- Optimized rendering for hundreds of interactive 3D components through asset optimization, memoization and lazy loading, keeping interaction smooth.
- Delivered **SHERPA**, a rack-level power and utilization dashboard, and **OASIS**, a 2D/3D floor-plan alarm and spatial search app.
- Built an **IoT-connected server room twin** with ESP32 telemetry, CFD-based airflow visualization and temperature overlays.
- Prototyped **NVIDIA Omniverse** digital-twin workflows with Kit SDK, OpenUSD scene composition and DSX Blueprint, and assessed interoperability with the web stack.
- Worked with UI/UX, backend and product teams to take features from design to production.

PROJECTS

Badvisor

- Building a **web-based 3D editor and viewer** in **Three.js and React Three Fiber**, focused on fast rendering and a clean, maintainable architecture.
- Buiding a **camera system** with orbit and first-person cameras: inertia, auto-rotation, zoom and pan sensitivity, angle and distance limits, hover-parallax effects, per-device focal lengths, and click-to-walk navigation with collision detection.
- Implementing **WebGPU rendering with automatic WebGL fallback**, plus a post-processing effects pipeline.

DCOS – Data Center Operating System (Hitachi Datacenter Simulation Platform)
Interactive 3D Digital Twin | React.js | Three.js | React Three Fiber | Croquet | WebSockets
- Developed an interactive 3D Digital Twin platform using React.js and Three.js for designing, visualizing, and managing enterprise data center infrastructure.
- Implemented drag-and-drop rack placement, responsive 3D layouts, and real-time infrastructure editing with grid-based spatial logic.
- Integrated glTF/GLB assets with React Three Fiber, implementing optimized materials, textures, lighting, camera controls, and rendering techniques for smooth browser performance.
- Integrated REST APIs and WebSockets for live monitoring of power, temperature, and infrastructure metrics with Croquet-based multi-user collaboration.
- Optimized rendering performance to efficiently manage hundreds of interactive 3D components with smooth user interaction.

SHERPA – Smart Energy & Power Allocation Platform
Real-Time Monitoring Dashboard | React.js | Highcharts | REST APIs | WebSockets
- Developed responsive React.js dashboards for monitoring rack-level power consumption, infrastructure utilization, and operational metrics.
- Integrated REST APIs and WebSockets for continuous live data streaming with Highcharts-based visualization.
- Designed reusable UI components supporting responsive layouts and high-frequency real-time updates.

IoT-Integrated 3D Digital Twin
Three.js | React Three Fiber | WebGL | ESP32 | WebSockets
- Developed a WebGL-based 3D server room simulation using Three.js, React Three Fiber, and glTF/GLB assets with live IoT telemetry integration.
- Implemented CFD-based airflow visualization, temperature overlays, and infrastructure monitoring dashboards.
- Enabled synchronized multi-user collaboration using Croquet while optimizing rendering performance for real-time interaction.

OASIS – Operational Alarm & Spatial Insights Search
React.js | Three.js | REST APIs
- Developed a responsive React.js application featuring interactive 2D/3D floorplan visualization and real-time infrastructure monitoring.
- Integrated REST APIs for operational alarms and infrastructure data with reusable UI components and spatial visualization modules.
- Improved infrastructure visibility through responsive layouts and seamless user interactions.

NVIDIA Omniverse Digital Twin
NVIDIA Omniverse | OpenUSD | Kit SDK | Python | Digital Twin
- Developed proof-of-concept Digital Twin workflows using NVIDIA Omniverse Kit and OpenUSD.
- Implemented USD scene composition, asset loading, and streaming workflows while evaluating enterprise Digital Twin capabilities using NVIDIA Omniverse.
- Configured Omniverse Kit applications and DSX Blueprint to visualize industrial assets and evaluate Digital Twin capabilities.
- Evaluated interoperability between Omniverse, OpenUSD, and web-based visualization platforms for future enterprise deployment.

TECHNICAL SKILLS

Frontend : React.js • JavaScript (ES6+) • TypeScript • HTML5 • CSS3 • Tailwind CSS • Redux
3D Technologies: Three.js • React Three Fiber (R3F) • WebGL • glTF/GLB • Croquet • Highcharts • D3.js • OpenUSD • NVIDIA Omniverse
APIs & Real-Time : REST APIs • WebSockets • Socket.IO • JSON
Backend : Node.js • Express.js
Databases : MongoDB • MySQL
Tools : Git • GitHub • VS Code • Postman • Vercel • Netlify • Render